

# Apriori Unimodality: Attributing polarization to sociodemographic groups

Dimitris Tsirmpas, John Pavlopoulos

March 2025

## 1 Introduction

Annotations are essential for a wide range of tasks, especially in fields such as content moderation, toxicity detection and hate speech detection. These tasks are especially essential for training systems that protect vulnerable and minority groups in online spaces, where they are often targeted [17, 4]. However, annotations for such tasks are subjective, and are usually based on majority-group annotators. This may limit the usefulness of the data, or even lead to biased and misconfigured systems.

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree in vain. An example of such cases would be racist comments that are presented with coded language (usually referred to as “dog-whistles” [13]), which are not picked up by most annotators, but only by ones belonging in the targeted minority. Inversely, majority-group annotators may bias their own annotations against the minority-group annotators; CITATION for instance shows that toxicity models disproportionately marked African American speech as “toxic”. This issue exists even in representative samples.

*Inter-annotator agreement* is often used to detect such cases [8]. Metrics based on agreement mostly measure annotation variance, which may not be an optimal problem formulation for detecting such cases. *Polarization* is a better instrument in this regard, since it formulates annotation analysis as a cluster identification problem [11, 12]—specifically whether two or more annotation clusters are present. Figure 1

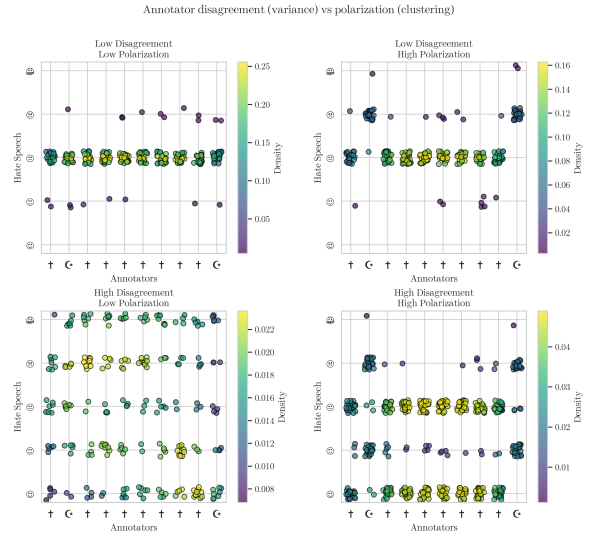


Figure 1: Disagreement is similar to variance; thus it overlooks cases where annotators may be “split down the middle”. Polarization on the other hand, is able to identify multiple clusters, even if they are close together.

displays how the two methods handle four common cases of annotation distributions for a hypothetical toxicity annotation task.

While we can detect polarization [12, 11], there is little work on how to detect whether it is actually caused by marginalized groups. Pavlopoulos and Likas [12] propose a mechanism which only works in the extreme case where overall polarization is zero; in this work we demonstrate that this case is not fre-

quent in real-world data. We instead propose the “*Apunim Metric*”, which attributes polarization to individual annotator sociodemographic groups. We provide a formulation which avoids inherent statistical limitations. Our contributions are:

1. We introduce a new metric which attributes polarization to specific sociodemographic groups.
2. We discover issues not yet acknowledged in literature, such as the presence of inherent polarization and conflicting polarization directions. Our proposed metric is designed to avoid these issues.
3. We apply our test to four datasets; two with human comments and annotations [15, 9] as well as two generated and annotated by Large Language Model (LLM) agents [16].

## 2 Methodology

In this section, we provide a formal mathematical formulation for the problem of attributing polarization to specific annotator characteristics (§2.1), and offer an intuitive rationale for how established polarization metrics can be leveraged (§2.2). We then introduce a data-point-level statistic that attributes polarization to individual SocioDemographic Background (SDB) groups (§2.3), and subsequently develop a statistic that generalizes this mechanism to an entire dataset.

### 2.1 Problem Formulation

**SocioDemographic Backgrounds** Since our goal is to pinpoint which specific characteristics contribute to polarization, we need to isolate individual groups within a SDB. Let  $\Theta$  be one of multiple “dimensions” the set of all annotator SDB groups for a single dimension (e.g. “male”, “female” and “non-binary” if we are investigating annotator gender).

**Annotations** Now let  $d = \{c_1, c_2, \dots\}$  be a dataset composed of multiple annotated data points. We assume that annotating a data point depends on two variables: inherent (“apriori”) controversy, and

the (“aposteriori”) annotator’s SDB, as well as uncontrolled factors such as mood and personal experiences (noise). Assuming that each data-point  $c$  is assigned multiple annotators, we can define the annotation multi-set  $A(c) = \{(a, \theta)\}$ , where  $a$  is a single annotation. As an example, the annotations for a comment in a sentiment analysis task would be formulated as:

$$A(\text{"Could be better, could be worse"}) = \{(\text{positive}, \text{female}), (\text{negative}, \text{male}), (\text{positive}, \text{female}), \dots\} \quad (1)$$

**Polarization** Each set of annotations features a certain degree of *polarization*. Unpolarized annotations are usually unimodal; the annotators disagree on the details (which would be shown roughly as a bell curve around the median annotation), not fundamentally (which would likely be shown as a multimodal distribution). Pavlopoulos and Likas [12] create a polarization metric, the “normalized Distance From Unimodality ( $nDFU$ )”, which measures whether an annotation set shows no polarization (unimodal distribution —  $nDFU \rightarrow 0$ ), up to complete polarization (multimodal distribution —  $nDFU \rightarrow 1$ ). Thus, we need a metric that measures how much the  $nDFU$  of a data point can be partially explained by  $\theta$  (given that polarization exists).

### 2.2 Quantifying changes in polarization

**Single data point polarization** Intuitively,  $\theta$  partially explains the polarization of a data point  $c$  when the annotations grouped by  $\theta$  are more polarized compared to the full set of annotations. Figure 3 exhibits a hypothetical example where a misogynistic comment is annotated for toxicity by male and female annotators.<sup>1</sup> The annotations are generally polarized ( $nDFU_{all} = 0.625$ ). The annotations by female annotators exhibit low polarization between themselves ( $nDFU_{women} = 0.1$ ), since most agree the data-point

<sup>1</sup>The use of only two predominant genders is made only for the purposes of demonstration.

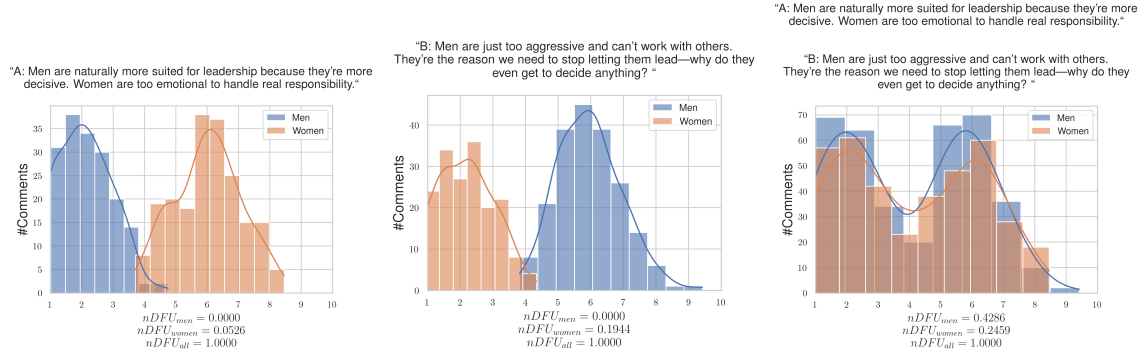


Figure 2: Hypothetical example of a polarizing discussion with two comments, for which the annotators disagree on which is the toxic one. If we aggregate the two comments, the polarization scores for both men and women significantly rise, obscuring whether the exhibited polarization can be partially attributed to gender.

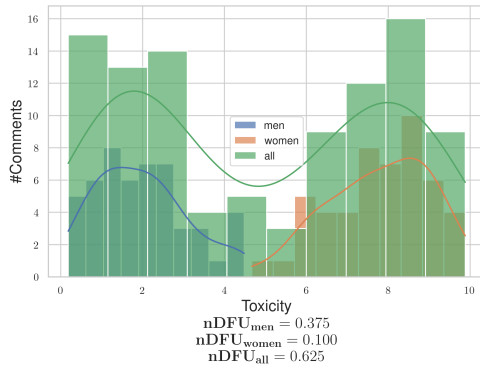


Figure 3: Hypothetical example of a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender. Overall polarization is much greater than the polarization exhibited by the annotations grouped by gender.

is toxic. The set of male annotations, also shows low polarization ( $nDFU_{men} = 0.375$ ), but for the opposite reason—most men agree that it is *not* toxic. This suggests that the overall polarization is driven by disagreements between male and female annotators.

**Dataset polarization** Given this observation, we would be tempted to aggregate all annotations in the dataset and compare the polarization of each  $\theta$  with the full set of annotations  $A(c)$ . In fact this is very tempting since aggregating annotations solves the very real and important problem of annotation scarcity; each data point may be annotated by only 5 annotators, leading to a best-case 3-2 split between groups. Since  $nDFU$  is at its core a histogram estimation, 3 annotations may not be enough to acquire a reliable histogram (this is discussed in depth in § 2.6).

However, this formulation would not work well. Figure 2 illustrates a hypothetical discussion with two comments, both of which are toxic, but where male and female annotators disagree on *which* comment is the toxic one. If we aggregate the annotations to a single discussion, the opposing polarization effects might balance each other out, leading to a false negative. This example demonstrates that polarization has a *direction*, which is not captured by the  $nDFU$  metric which is symmetric. Thus, care must be taken to not combine polarization effects of opposite directions.

## 2.3 Observed polarization

As demonstrated above, it is necessary to define a metric for the polarization attribution of each individual data point when grouped by  $\theta$ . We thus introduce the “*pol-statistic*”, which quantifies the change in  $nDFU$  when annotations are grouped by  $\theta \in \Theta$ . The observed polarization of a data-point on the group of annotations  $\theta$  can be directly computed as:

$$pol(c, \theta) = nDFU(P(A(c), \theta)) \quad (2)$$

where  $P(A(c), \theta_i) = \{a(c; \theta) \in A(c) | \theta = \theta_i\}$  is the partition of  $A$  for a data-point  $c$ , for which its annotations belong to the SDB group  $\theta_i$ .

## 2.4 Apriori polarization

In § 1 we mentioned that annotator polarization can be caused either by the annotators SDB, or other factors. The latter can be considered “apriori” polarization, which is driven by the inherent “controversy” of the data point. Previous work [12] considered the case where this apriori polarization was zero; this is not the case in many datasets, as we demonstrate in this section.

In § 2.2 we compared the polarization of the whole set of annotations with the partitions by annotator group. This is *not statistically sound*, since we are comparing statistics of a subset with its superset, violating the fundamental hypothesis of dependent samples. When  $A$  is a proper subset of  $S$ , the data points composing  $A$  are necessarily included in  $S$ . This inclusion induces an intrinsic, non-zero covariance between any statistic calculated on  $A$  (e.g., the sampled mean  $\bar{X}_A$ ) and the corresponding statistic calculated on  $S$  (e.g., the sampled mean  $\bar{X}_S$ ). We can not ignore this problem due to the extreme overlap exhibited by tiny annotation sets.

A standard method of side-stepping this issue is to compare the subset with its complement to the superset (comparing  $\bar{X}_A$  with  $\bar{X}_{S/A}$ ). This can not be applied to our case. Consider the example in Figure 3; if we compared  $nDFU_{men}$  with  $nDFU_{women}$ , the result would be very close to zero, indicating no polarization attribution to gender.

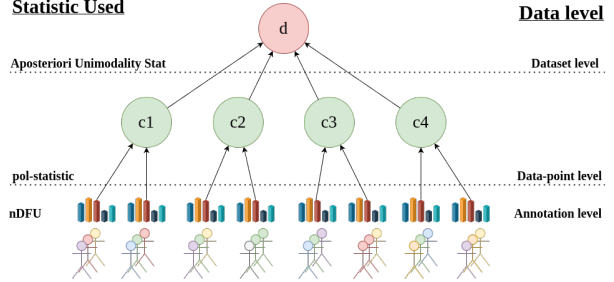


Figure 4: An overview of the Aposteriori Unimodality Test. We gather statistical information from the annotations (different SDBs are denoted by different colors) via the  $nDFU$  measure, which is aggregated on the data-point-level by the *pol-statistic*. The Aposteriori Unimodality Test is applied on the discussion level, operating on the *pol-statistics* of the individual data-points.

Given infinite annotations, the assumption of independence would work. Likewise, if we compared  $A$  with infinite random subsets of  $S$ . We can thus estimate the apriori polarization in data point  $c$  by bootstrapping. We randomly partition the annotations in  $|\Theta|$  groups, with matching group sizes (e.g., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we will create random partitions of sizes 80 and 20). The randomly partitioned annotations are then sampled  $t$  times. Formally, the estimated apriori polarization will be given by:

$$\frac{1}{t} \sum_{i=1}^t nDFU(\tilde{P}_i(A(c))) \quad (3)$$

where  $\tilde{P}_i$  is the random partition operator with matching group sizes.

## 2.5 Aposteriori polarization

While data scarcity is an important limitation when acquiring the observed polarization, this is decisively not the case with the number of data points themselves, which can range up to hundreds of thousands. Thus, we have the luxury of filtering data points be-

fore computing a dataset-wide metric. In our case, data points which feature no polarization are useless in a metric that attributes polarization, as well as data points which are composed of only a single group. Thus, we define the subset  $S_d$  of dataset  $d$  such that:

$$S_d = \{c \in d \mid nDFU(c) > \alpha \text{ and } c \text{ annotated by at least two groups}\} \quad (4)$$

where  $\alpha$  is a small positive number (in our experiments  $\alpha = 0.2$ ).

By obtaining the *pol-statistics* for all data-points in  $S_d$  for SDB  $\theta$ , we can get the mean observed polarization for the dataset:

$$pol_{apost}(d, \theta) = \frac{1}{|d|} \sum_{c \in S_d} pol(c, \theta) \quad (5)$$

and similarly the mean apriori dataset polarization:

$$pol_{apriori}(d) = \frac{1}{|d|} \sum_{c \in S_d} \frac{1}{t} \sum_{i=1}^t nDFU(\tilde{P}_i(A(c))) \quad (6)$$

We can then define the *Apunim Statistic* as:

$$apunim(d, \theta) = \frac{pol_{apost}(d, \theta) - pol_{apriori}(d)}{1 - pol_{apriori}(d)} \quad (7)$$

If  $apunim(d, \theta) \approx 0$ , the exhibited polarization among annotators of group  $\theta$  does not surpass what would be expected by chance. If  $apunim(d, \theta) > 0$ , then the group partially explains a rise in polarization, and inversely for the case  $apunim(d, \theta) < 0$ . The scope of the statistic and its relationship with the previously presented statistics are demonstrated in Figure 4.

While it is possible to obtain p-value results for the *apunim* values being larger or less than 0 respectively, we find that the inherent variance of our statistic makes a p-value estimation worthless ( $p \gg 0.1$ ). Nevertheless, we include a definition for parametric and non-parametric p-value estimation, and our results in Appendix B.

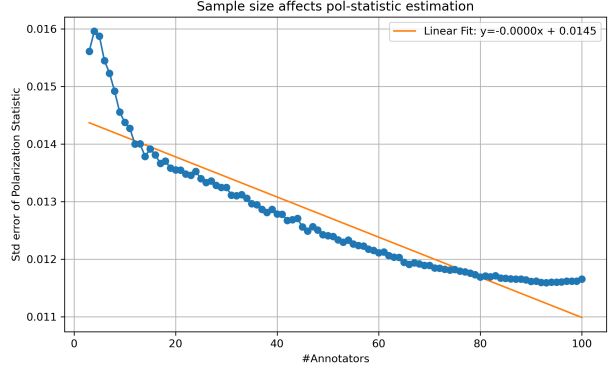


Figure 5: Standard error of the *pol-statistic* when sampled with replacement from the 100 LLM-annotator synthetic dataset for various number of annotators.

## 2.6 Reliability of the pol statistic

In §2.5 we mentioned that the test relies on enough annotations for each SDB group and data-point. This can be an issue, given the cost required to utilize multiple human annotators for each data-point [14].

We use a subset of synthetic discussions from a synthetic dataset, the Virtual Moderation Dataset (VMD) [16]. We use 100 LLM annotators with different SDB characteristics and sample progressively 3-100 annotators with replacement, then calculate the standard error with the mean *pol-statistic*. As expected, the standard error is inversely proportional to the number of annotators, although the difference is not great, even for a low number of annotators.

Figure 5 demonstrates the effect of the number of annotators to the *pol-statistic* estimation. We can see that the *pol statistic* becomes consistent with about 15 annotators per comment. After about 80 annotators, we start to get diminishing results. This highlights a core limitation of the *apunim* metric.

## 3 Datasets

We use the datasets provided by Kumar et al. [9] and Sap et al. [15]. The datasets feature various online comments, each annotated by a number of hu-

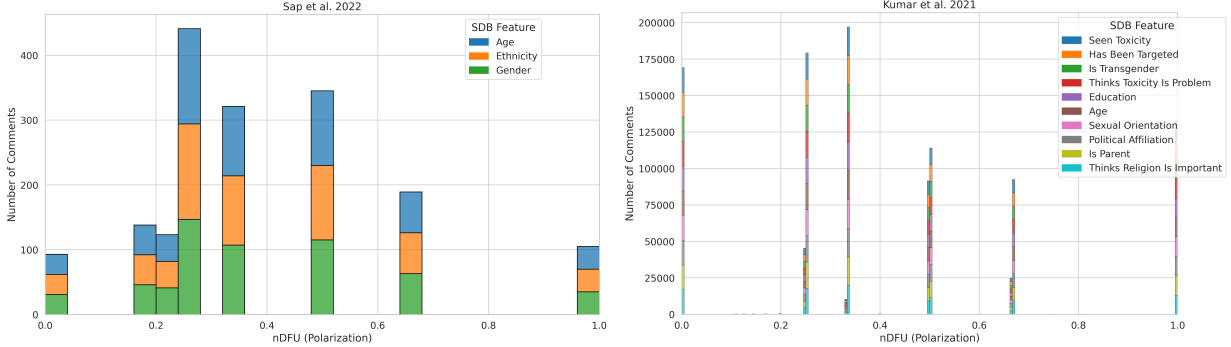


Figure 6: Histogram of dataset polarization for the human datasets considered in this paper per SDB feature.

man annotators (5 and 4-6 annotators per comment respectively). There is a reasonable amount of polarization in most annotations for both dataset, as demonstrated by Figure 6.

The Sap et al. [15] dataset includes racism annotations for 626 Twitter/X comments, and standard SDB information (age, race, education, gender). The Kumar et al. [9] dataset measures toxicity on various online comments, and the provided SDBs include mostly annotator experiences (e.g., whether they have been personally targeted online) as well as standard SDB information such as sexual orientation, age and education. Since the dataset is extensive (106035 comments), we only select a sample of 10000 comments, due both to computational constraints.

## 4 Results

Assuming a confidence level of  $\alpha = 0.1$  we test whether any of the provided SDB dimensions can partially explain polarization in the toxicity/racism annotations. The results for the Kumar et al. [9] and Sap et al. [15] datasets can be found in Tables ??, ?? respectively.

## 5 Conclusion

We introduced the ‘‘Aposteriori Unimodality Test’’, a statistical test that detects whether certain sociode-

Table 1: Apunim results for the Sap et al. 2022 dataset

| SDB Feature |                | apunim  |
|-------------|----------------|---------|
| Age         | (-0.001, 20.0] | 0.0000  |
|             | (20.0, 40.0]   | 0.0017  |
|             | (40.0, 60.0]   | 0.0115  |
|             | (60.0, 80.0]   | 0.0049  |
| Ethnicity   | black          | -0.2967 |
|             | hisp           | 0.0000  |
|             | middleEastern  | 0.0000  |
|             | other          | 0.0000  |
|             | white          | 0.1400  |
| Gender      | man            | 0.0942  |
|             | nonBinary      | 0.0000  |
|             | woman          | -0.1810 |

mographic groups partly cause polarization in annotations. Our test is resistant to low samples of annotations, bypasses common statistical issues such as sample dependence, and avoids newly-discovered issues such as the presence of inherent polarization and polarization direction. We apply our test two human-generated and two LLM-generated datasets and find interesting patterns on the former, and verify current literature on LLM SDB prompting on the latter.

## 6 Limitations

Table 2: Aposteriori Unimodality kappa and pvalue results for the Kumar et al. 2021 dataset

| SDB Feature    |                           |           |
|----------------|---------------------------|-----------|
| Age            | 18 - 24                   | -0.030410 |
|                | 25 - 34                   | 0.040010  |
|                | 35 - 44                   | -0.010029 |
|                | 45 - 54                   | 0.018776  |
|                | 55 - 64                   | 0.003176  |
|                | 65 or older               | 0.017648  |
|                | Prefer not to say         | 0.000000  |
| Education      | Associate degree          | -0.055727 |
|                | Bachelor's degree         | 0.095408  |
|                | College, no degree        | -0.108984 |
|                | Doctoral degree           | 0.008145  |
|                | High School graduate      | -0.040907 |
|                | Master's degree           | 0.054708  |
|                | No high school            | -0.007941 |
|                | Other                     | 0.000000  |
|                | Prefer not to say         | 0.000000  |
| Targeted       | False                     | -0.095286 |
|                | True                      | 0.112036  |
| Is Parent      | No                        | -0.136264 |
|                | Prefer not to say         | -0.004510 |
|                | Yes                       | 0.093870  |
| Is Transgender | No                        | -0.151712 |
|                | Prefer not to say         | 0.515615  |
|                | Yes                       | 0.016393  |
| Politics       | Conservative              | 0.066361  |
|                | Independent               | -0.016543 |
|                | Liberal                   | -0.039058 |
|                | Other                     | -0.018498 |
|                | Prefer not to say         | -0.001853 |
| Seen Toxicity  | False                     | 0.145958  |
|                | True                      | -0.117346 |
| Sexual Orient. | Bisexual                  | 0.179856  |
|                | Heterosexual              | -0.131972 |
|                | Homosexual                | -0.044238 |
|                | Other                     | 0.003322  |
|                | Prefer not to say         | -0.028099 |
|                | NaN                       | 0.077491  |
| Religion       | Not important             | -0.167580 |
|                | Not too important         | -0.026466 |
|                | Prefer not to say         | 0.004367  |
|                | Somewhat important        | 0.036322  |
|                | Very important            | 0.088742  |
| Toxicity       | Frequently a problem      | -0.057269 |
|                | Not a problem             | 0.039260  |
|                | Occasionally a problem    | -0.009915 |
|                | Rarely a problem          | 0.044619  |
|                | Very frequently a problem | -0.057531 |

Since there are no other established tests that attribute polarization to sociodemographic characteristics, there is no way to verify that the results presented in §4 are accurate. Similarly, there is no qualitative way of evaluating our test, other than observations on the (non-) efficacy of LLM SDB prompting and our own intuition. Lastly, while our test seems to work sufficiently well with a small number of annotators per data-point, we still encourage researchers using this test to have at least a few annotators of each sociodemographic group that is being tested, in their dataset.

## 7 Ethical Considerations

While our works aims to help protect marginalized and disadvantaged groups by attributing polarization to certain subgroups, it can be taken advantage of by malicious actors. Given a dataset with fine-grained SDB information, these actors can use our test to target specific vulnerable groups. We thus urge researchers to keep datasets including such information protected, and provided to others only under explicit permission.

## References

- Jacy Reese Anthis et al. *LLM Social Simulations Are a Promising Research Method*. 2025. arXiv: 2504.02234 [cs.HC]. URL: <https://arxiv.org/abs/2504.02234>.
- James Bisbee et al. “Synthetic Replacements for Human Survey Data? The Perils of Large Language Models”. In: *Political Analysis* 32.4 (2024), 401–416. DOI: 10.1017/pan.2024.5.
- S.Y. Chen, Z. Feng, and X. Yi. “A general introduction to adjustment for multiple comparisons”. In: *Journal of Thoracic Disease* 9.6 (2017). doi: 10.21037/jtd.2017.05.34; PMID: 28740688; PMCID: PMC5506159, pp. 1725–1729.

- [4] Shelby Dioum. *What Explains the Increase in Online Hate Speech?* Accessed: 2025-09-22. University of California, Davis, Mar. 2025. URL: <https://www.ucdavis.edu/magazine/what-explains-increase-online-hate-speech>.
- [5] Luke Hewitt et al. “Predicting Results of Social Science Experiments Using Large Language Models”. Equal contribution, order randomized. Aug. 2024.
- [6] Sture Holm. “A Simple Sequentially Rejective Multiple Test Procedure”. In: *Scandinavian Journal of Statistics* 6.2 (1979), pp. 65–70. ISSN: 03036898, 14679469. URL: <http://www.jstor.org/stable/4615733> (visited on 07/07/2025).
- [7] Bernard J. Jansen, Soon gyo Jung, and Joni Salminen. “Employing large language models in survey research”. In: *Natural Language Processing Journal* 4 (2023), p. 100020. ISSN: 2949-7191. DOI: <https://doi.org/10.1016/j.nlp.2023.100020>. URL: <https://www.sciencedirect.com/science/article/pii/S2949719123000171>.
- [8] Petra Kralj Novak et al. “Handling Disagreement in Hate Speech Modelling”. In: *Information Processing and Management of Uncertainty in Knowledge-Based Systems*. Ed. by Davide Ciucci et al. Cham: Springer International Publishing, 2022, pp. 681–695. ISBN: 978-3-031-08974-9.
- [9] Deepak Kumar et al. “Designing toxic content classification for a diversity of perspectives”. In: *Proceedings of the Seventeenth USENIX Conference on Usable Privacy and Security*. SOUPS’21. USA: USENIX Association, 2021. ISBN: 978-1-939133-25-0.
- [10] Terrence Neumann, Maria De-Arteaga, and Sina Fazelpour. *Should you use LLMs to simulate opinions? Quality checks for early-stage deliberation*. 2025. arXiv: 2504.08954 [cs.CY]. URL: <https://arxiv.org/abs/2504.08954>.
- [11] John Pavlopoulos and Aristidis Likas. “Distance from Unimodality for the Assessment of Opinion Polarization”. In: *Cognitive Computation* 15.2 (2023), pp. 731–738. ISSN: 1866-9964. DOI: 10.1007/s12559-022-10088-2. URL: <https://doi.org/10.1007/s12559-022-10088-2>.
- [12] John Pavlopoulos and Aristidis Likas. “Polarized Opinion Detection Improves the Detection of Toxic Language”. In: *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Yvette Graham and Matthew Purver. St. Julian’s, Malta: Association for Computational Linguistics, Mar. 2024, pp. 1946–1958. URL: <https://aclanthology.org/2024.eacl-long.117/>.
- [13] Anne Quaranto. “Dog whistles, covertly coded speech, and the practices that enable them”. In: *Synthese* 200.4 (2022), p. 330.
- [14] Luca Rossi, Katherine Harrison, and Irina Shklovski. “The Problems of LLM-generated Data in Social Science Research”. In: *Sociologica* 18.2 (2024), 145–168. DOI: 10.6092/issn.1971-8853/19576. URL: <https://sociologica.unibo.it/article/view/19576>.
- [15] Maarten Sap et al. “Annotators with Attitudes: How Annotator Beliefs And Identities Bias Toxic Language Detection”. In: *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Ed. by Marine Carpuat, Marie-Catherine de Marneffe, and Ivan Vladimir Meza Ruiz. Seattle, United States: Association for Computational Linguistics, July 2022, pp. 5884–5906. DOI: 10.18653/v1/2022.naacl-main.431. URL: <https://aclanthology.org/2022.naacl-main.431/>.
- [16] Dimitris Tsirmpas, Ion Androutsopoulos, and John Pavlopoulos. *Scalable Evaluation of Online Facilitation Strategies via Synthetic Simulation of Discussions*. 2025. arXiv: 2503.16505

[cs.HC]. URL: <https://arxiv.org/abs/2503.16505>.

- [17] United Nations. *Targets of Hate*. Accessed: 2025-09-22. Sept. 2025. URL: <https://www.un.org/en/hate-speech/impact-and-prevention/targets-of-hate>.

## A Acronyms

**VMD** Virtual Moderation Dataset

**LLM** Large Language Model

**SDB** SocioDemographic Background

**nDFU** normalized Distance From Unimodality

**FWER** Family-Wise Error Rate

## B p-value Estimation

### B.1 Methodology

Like most metrics, we also include a p-value alongside the *apunim* value. This can be computed either non-parametrically, by repeatedly shuffling annotations and computing a null distribution of *apunim*, or parametrically, by assuming a normal distribution (which is generally a safe assumption if  $t \geq 30$ ). In the latter case, we compute:

$$z = \frac{\hat{\kappa} - \mathbb{E}[\kappa]}{SE(\hat{\kappa})}, p = 1 - \Phi(z) \quad (8)$$

Since we are simultaneously considering  $|\Theta|$  hypotheses, we apply a multiple comparison correction to the resulting p-values. We choose the Holms method [6]. The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [3]). In general, it is safe to set FWER equal to the significance level of our test (e.g.,  $FWER = 0.95$  if we are looking for  $p < 0.05$ ).

## B.2 Results

The results for the datasets presented in § 3 can be found in Tables ??, ?? for the Sap et al. [15] and Kumar et al. [9] datasets respectively. The experiments were ran on the exact same parameters as in § 4.

Note the absence of any statistically significant results for both parametric and non-parametric tests. This suggests that, although the differences in polarization captured by the *pol* statistic introduced in § 2 are substantial enough to hint at polarization being attributable to certain groups, they remain too small to overcome the inherent noise in human annotations. As a result, we cannot obtain a robust estimate of statistical significance.

## C Synthetic Datasets

We use two synthetic datasets; one is the VMD presented in Tsirmpas, Androutsopoulos, and Pavlopoulos [16]. This dataset features 140 discussions, each having 14 comments (and usually up to 14 facilitator comments), each comment annotated by 10 LLM annotators, each supplied with a different SDB. The second dataset, is an extension of VMD, where we select four random unmoderated discussions, and employ 100 distinct LLM annotators.

An issue with synthetic datasets is the lack of overall polarization (see Figure 7). This can be offset by the ability of synthetic datasets to be scaled to a much greater extent than human datasets. This is because LLMs do not have cost and availability constraints of humans.

The results for the 100-annotator dataset can be found in Table ??. We can not obtain results for the VMD dataset, since no comment exhibited a priori polarization while also being represented by more than two annotators of different groups. The lack of overall polarization, as well as the quantitatively smaller *apunim* values, suggest that SDB prompting can not be used to simulate annotations for human groups. This is consistent with relevant literature on the topic [1, 5, 14, 7, 2, 10].

Table 3: Apunim results for the sap dataset

| SDB Feature |                | apunim  | p_param | p_nonparam |
|-------------|----------------|---------|---------|------------|
| Age         | (-0.001, 20.0] | 0.0000  | NaN     | 1.0000     |
|             | (20.0, 40.0]   | 0.0017  | 0.9958  | 0.9977     |
|             | (40.0, 60.0]   | 0.0115  | 0.9794  | 0.9976     |
|             | (60.0, 80.0]   | 0.0049  | 0.9660  | 1.0000     |
| Ethnicity   | black          | -0.2967 | 0.6369  | 0.6564     |
|             | hisp           | 0.0000  | NaN     | 1.0000     |
|             | middleEastern  | 0.0000  | NaN     | NaN        |
|             | other          | 0.0000  | NaN     | 1.0000     |
|             | white          | 0.1400  | 0.7130  | 0.8246     |
| Gender      | man            | 0.0942  | 0.8368  | 0.8920     |
|             | nonBinary      | 0.0000  | NaN     | 1.0000     |
|             | woman          | -0.1810 | 0.7198  | 0.7653     |

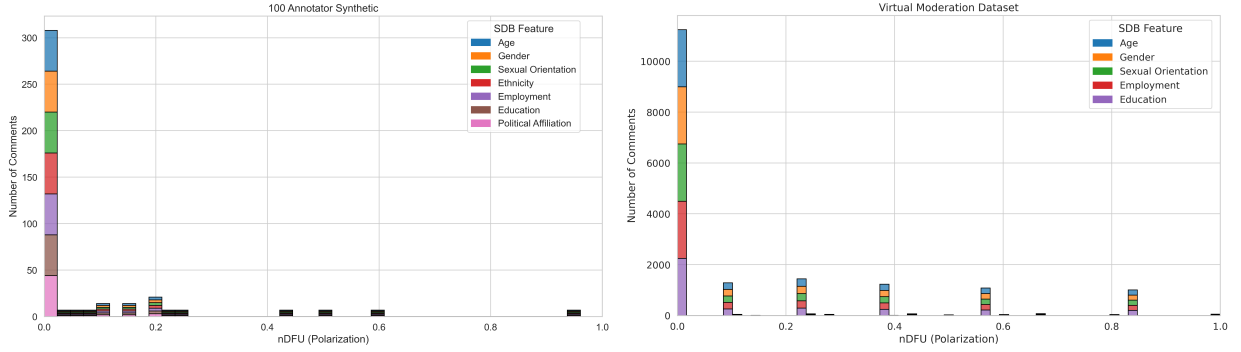


Figure 7: Histogram of dataset polarization for the synthetic datasets considered in this paper per SDB feature.

Table 4: Aposteriori Unimodality kappa and pvalue results for the Kumar et al. 2021 dataset

| SDB Feature                  |                           | apunim    | pvalue_parametric | pvalue_nonparametric |
|------------------------------|---------------------------|-----------|-------------------|----------------------|
| Age                          | 18 - 24                   | -0.030410 | 0.954856          | 0.967112             |
|                              | 25 - 34                   | 0.040010  | 0.948614          | 0.963320             |
|                              | 35 - 44                   | -0.010025 | 0.987850          | 0.998125             |
|                              | 45 - 54                   | 0.018776  | 0.973235          | 0.971674             |
|                              | 55 - 64                   | 0.003176  | 0.994370          | 1.000000             |
|                              | 65 or older               | 0.017648  | 0.959945          | 1.000000             |
|                              | Prefer not to say         | 0.000000  | NaN               | 1.000000             |
| Education                    | Associate degree          | -0.055727 | 0.923705          | 0.947000             |
|                              | Bachelor's degree         | 0.095408  | 0.878875          | 0.908507             |
|                              | College, no degree        | -0.108984 | 0.862214          | 0.927355             |
|                              | Doctoral degree           | 0.008115  | 0.953037          | 1.000000             |
|                              | High School graduate      | -0.040907 | 0.937773          | 0.980510             |
|                              | Master's degree           | 0.054708  | 0.935074          | 0.978080             |
|                              | No high school            | -0.007941 | 0.978720          | 1.000000             |
|                              | Other                     | 0.000000  | NaN               | 1.000000             |
|                              | Prefer not to say         | 0.000000  | NaN               | 1.000000             |
| Has Been Targeted            | Professional degree       | 0.001086  | 0.994127          | 1.000000             |
|                              | False                     | -0.095286 | 0.851124          | 0.840822             |
| Is Parent                    | True                      | 0.112036  | 0.838550          | 0.851233             |
|                              | No                        | -0.136264 | 0.793274          | 0.834386             |
|                              | Prefer not to say         | -0.004510 | 0.983443          | 1.000000             |
| Is Transgender               | Yes                       | 0.093870  | 0.851837          | 0.852657             |
|                              | No                        | -0.151712 | 0.796024          | 0.776316             |
|                              | Prefer not to say         | 0.515608  | 0.356331          | 0.416667             |
| Political Affiliation        | Yes                       | 0.016393  | 0.973112          | 0.969231             |
|                              | Conservative              | 0.066361  | 0.918828          | 0.974428             |
|                              | Independent               | -0.016543 | 0.979319          | 0.992056             |
|                              | Liberal                   | -0.039058 | 0.951880          | 0.977920             |
|                              | Other                     | -0.018498 | 0.949256          | 1.000000             |
| Seen Toxicity                | Prefer not to say         | -0.001853 | 0.996053          | 1.000000             |
|                              | False                     | 0.145958  | 0.789892          | 0.810560             |
| Sexual Orientation           | True                      | -0.117346 | 0.816276          | 0.831252             |
|                              | Bisexual                  | 0.179856  | 0.786152          | 0.789989             |
|                              | Heterosexual              | -0.131972 | 0.808635          | 0.824752             |
|                              | Homosexual                | -0.044238 | 0.940537          | 0.940860             |
|                              | Other                     | 0.003322  | 0.979759          | 1.000000             |
|                              | Prefer not to say         | -0.028099 | 0.957889          | 1.000000             |
| Thinks Religion Is Important | NaN                       | 0.077491  | 0.756128          | 1.000000             |
|                              | Not important             | -0.167580 | 0.790175          | 0.858974             |
|                              | Not too important         | -0.026466 | 0.961419          | 0.982239             |
|                              | Prefer not to say         | 0.004367  | 0.983731          | 1.000000             |
|                              | Somewhat important        | 0.036322  | 0.954503          | 0.975619             |
| Thinks Toxicity Is Problem   | Very important            | 0.088742  | 0.893641          | 0.964399             |
|                              | Frequently a problem      | -0.057269 | 0.927206          | 0.983830             |
|                              | Not a problem             | 0.039260  | 0.934240          | 0.997835             |
|                              | Occasionally a problem    | -0.009915 | 0.987592          | 0.996255             |
|                              | Rarely a problem          | 0.044619  | 0.943330          | 0.956728             |
|                              | Very frequently a problem | -0.057531 | 0.914521          | 0.954014             |

Table 5: Apunim results for the virtual dataset

| SDB Feature        |                      | apunim  | p_param | p_nonparam |
|--------------------|----------------------|---------|---------|------------|
| Age                | (20.956, 32.0]       | -0.0114 | 0.9690  | 0.9815     |
|                    | (32.0, 43.0]         | -0.0312 | 0.8992  | 0.9446     |
|                    | (43.0, 54.0]         | -0.0094 | 0.9698  | 0.9800     |
|                    | (54.0, 65.0]         | 0.0180  | 0.9103  | 0.9469     |
| Education          | Bachelor's           | -0.0103 | 0.9670  | 0.9838     |
|                    | Master's             | 0.0129  | 0.9647  | 0.9769     |
|                    | No formal education  | 0.0000  | NaN     | 1.0000     |
|                    | PhD                  | 0.0258  | 0.9161  | 0.9554     |
|                    | Some College         | -0.0296 | 0.9035  | 0.9562     |
| Employment         | Botanist             | 0.0000  | NaN     | 1.0000     |
|                    | Cybersecurity Expert | 0.0000  | NaN     | 1.0000     |
|                    | Farmer               | 0.0000  | NaN     | 1.0000     |
|                    | Game Developer       | 0.0000  | NaN     | 1.0000     |
|                    | Historian            | 0.0000  | NaN     | 1.0000     |
|                    | Poet                 | 0.0000  | NaN     | 1.0000     |
|                    | Registered Nurse     | 0.0000  | NaN     | 1.0000     |
|                    | Research Scientist   | 0.0000  | NaN     | 1.0000     |
|                    | Retired Philosopher  | 0.0000  | NaN     | 1.0000     |
|                    | Stock Trader         | 0.0000  | NaN     | 1.0000     |
|                    | Travel Blogger       | 0.0000  | NaN     | 1.0000     |
| Gender             | female               | -0.0248 | 0.9325  | 0.9562     |
|                    | male                 | -0.0008 | 0.9983  | 0.9992     |
|                    | non-binary           | 0.0000  | NaN     | 1.0000     |
| Sexual Orientation | Asexual              | 0.0000  | NaN     | 1.0000     |
|                    | Bisexual             | 0.0000  | NaN     | 1.0000     |
|                    | Heterosexual         | 0.0170  | 0.9610  | 0.9638     |
|                    | Homosexual           | 0.0000  | NaN     | 1.0000     |
|                    | Lesbian              | 0.0000  | NaN     | 1.0000     |
|                    | Pansexual            | 0.0000  | NaN     | 1.0000     |